

Fixed Charge Transportation Problem

(and decomposition)

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Fixed charge transportation problem

Transport problem

- Bi-partite graph
- Sources $S_1, \dots, S_m$, terminals $T_1, \dots, T_n$
- Production s_i in S_i , demand d_j in T_j
- Edges have cost $c_{ij} \in \mathbb{R}$, infinite capacity

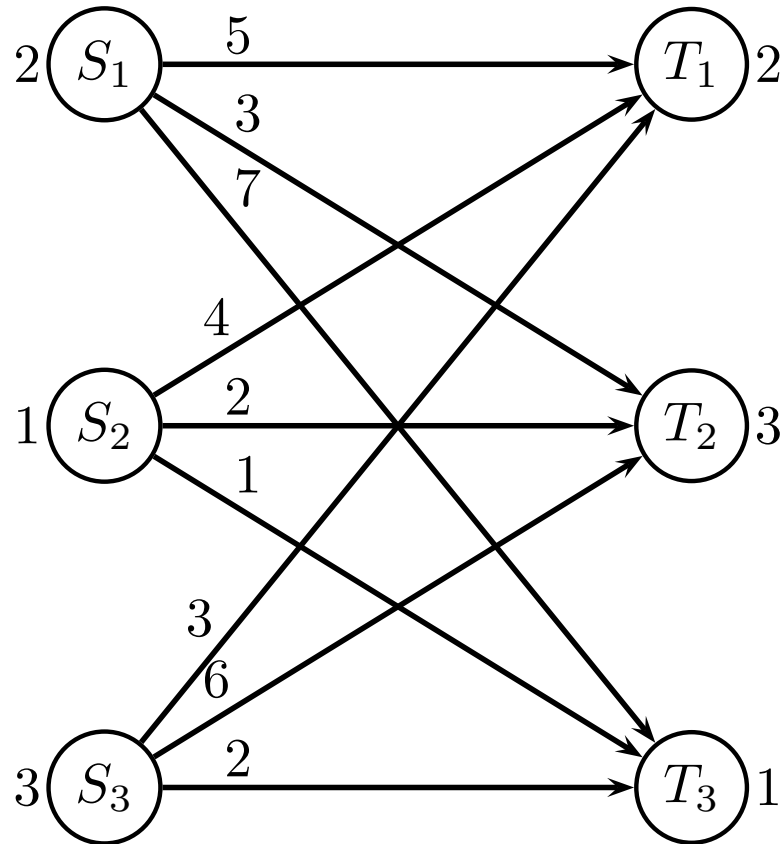
Now assume, **fixed charge** F for using an edge

- Cost for building connection (cable, pipeline)
- Startup cost for vehicle

If no flow on an edge, no fixed charge is paid

Fixed charge transportation problem

Example:



Fixed charge for using edge (i, j) is here $F = 8$
(could be different F_{ij} for each edge)

Formal definition

Without fixed cost

$$\begin{array}{ll}\min & \sum_i \sum_j c_{ij} x_{ij} \\ \text{s.t.} & - \sum_j x_{ij} = -s_i \quad \forall i = 1, \dots, m \\ & \sum_i x_{ij} = d_j \quad \forall j = 1, \dots, n \\ & x_{ij} \geq 0 \quad \forall i, j\end{array}$$

variable x_{ij} denotes flow from S_i to T_j

Formal definition

With fixed cost

$$\begin{array}{ll}\min & \sum_i \sum_j (c_{ij}x_{ij} + Fy_{ij}) \\ \text{s.t.} & -\sum_j x_{ij} = -s_i \quad \forall i = 1, \dots, m \\ & \sum_i x_{ij} = d_j \quad \forall j = 1, \dots, n \\ & x_{ij} \leq My_{ij} \quad \forall j = 1, \dots, n \\ & x_{ij} \geq 0 \quad \forall i, j \\ & y_{ij} \in \{0, 1\} \quad \forall i, j\end{array}$$

variable x_{ij} denotes flow from S_i to T_j

variable y_{ij} denotes whether edge from S_i to T_j is used

M is upper bound on flow

Fixed charge transportation problem

Variable x_{ij} denotes flow from S_i to T_j

Edge formulation *without* setup costs

$$\min 5x_{11} + 3x_{12} + 7x_{13} + 4x_{21} + 2x_{22} + 1x_{23} + 3x_{31} + 6x_{32} + 2x_{33}$$

$$\text{s.t. } -x_{11} \quad -x_{12} \quad -x_{13} \qquad \qquad \qquad = -2 \quad (u_1)$$

$$\qquad \qquad \qquad -x_{21} \quad -x_{22} \quad -x_{23} \qquad \qquad \qquad = -1 \quad (u_2)$$

$$\qquad \qquad \qquad \qquad \qquad \qquad -x_{31} \quad -x_{32} \quad -x_{33} = -3 \quad (u_3)$$

$$x_{11} \qquad \qquad \qquad +x_{21} \qquad \qquad \qquad +x_{31} \qquad \qquad \qquad = 2 \quad (v_1)$$

$$\qquad x_{12} \qquad \qquad \qquad +x_{22} \qquad \qquad \qquad +x_{32} \qquad \qquad \qquad = 3 \quad (v_2)$$

$$\qquad \qquad x_{13} \qquad \qquad \qquad +x_{23} \qquad \qquad \qquad +x_{33} = 1 \quad (v_3)$$

$$x_{11}, x_{12}, x_{13}, x_{21}, x_{22}, x_{23}, x_{31}, x_{32}, x_{33} \geq 0$$

Fixed charge transportation problem

Edge formulation *without* setup costs

```
min
  5 x11 + 3 x12 + 7 x13 + 4 x21 + 2 x22 + 1 x23 + 3 x31 + 6 x32 + 2 x33
subject to
- x11 - x12 - x13                                     = -2
               - x21 - x22 - x23                       = -1
                               - x31 - x32 - x33        = -3
x11               + x21               + x31            =  2
      x12          + x22               + x32            =  3
                x13          + x23               + x33   =  1
end
```

Fixed charge transportation problem

Edge formulation *with* setup costs (*)

```

min
  5 x11 + 3 x12 + 7 x13 + 4 x21 + 2 x22 + 1 x23 + 3 x31 + 6 x32 + 2 x33
+ 8 y11 + 8 y12 + 8 y13 + 8 y21 + 8 y22 + 8 y23 + 8 y31 + 8 y32 + 8 y33
subject to
- x11 - x12 - x13                                     = -2
               - x21 - x22 - x23                       = -1
                           - x31 - x32 - x33             = -3
x11               + x21               + x31             = 2
      x12               + x22               + x32         = 3
            x13               + x23               + x33   = 1

x11 - 3 y11 <= 0
x12 - 3 y12 <= 0
x13 - 3 y13 <= 0
x21 - 3 y21 <= 0
x22 - 3 y22 <= 0
x23 - 3 y23 <= 0
x31 - 3 y31 <= 0
x32 - 3 y32 <= 0
x33 - 3 y33 <= 0

binary
  y11 y12 y13 y21 y22 y23 y31 y32 y33
end

```

Much harder to solve (NP-hard)

Fixed charge transportation problem

IP-solution

MIP - Integer optimal solution: Objective = 4.8000000000e+01
Solution time = 0.00 sec. Iterations = 2 Nodes = 0

Variable Name	Solution Value
x12	2.000000
x22	1.000000
x31	2.000000
x33	1.000000
y12	1.000000
y22	1.000000
y31	1.000000
y33	1.000000

All other variables in the range 1-18 are 0.

Fixed charge transportation problem

LP-relaxed problem

Tried aggregator 1 time.

LP Presolve eliminated 9 rows and 0 columns.

Aggregator did 9 substitutions.

Reduced LP has 6 rows, 9 columns, and 18 nonzeros.

Presolve time = 0.00 sec.

Iteration log . . .

Iteration: 1 Dual objective = 14.000000

Dual simplex - Optimal: Objective = 3.2000000000e+01

Solution time = 0.00 sec. Iterations = 5 (0)

CPLEX> dis sol var -

Variable Name	Solution Value
x12	2.000000
x22	1.000000
x31	2.000000
x33	1.000000
y12	0.666667
y22	0.333333
y31	0.666667
y33	0.333333

All other variables in the range 1-18 are 0.

(bad bound, many fractional variables)

Fixed charge transportation problem

Problem

- Bound from LP problem is horribly bad. Often 50% from optimum
- Branch-and-bound algorithm will have huge running times
- Larger problems cannot be solved to optimality

Solution

- Reformulate problem (Dantzig-Wolfe decomposition)

Two-stage problems

Consider problem

$$\begin{array}{ll}\min & cx \\ \text{s.t.} & Ax = b\end{array}$$

Columns in A are result of a subproblem

$$(c_j, A_j) = \min\{dx : x \in S\}$$

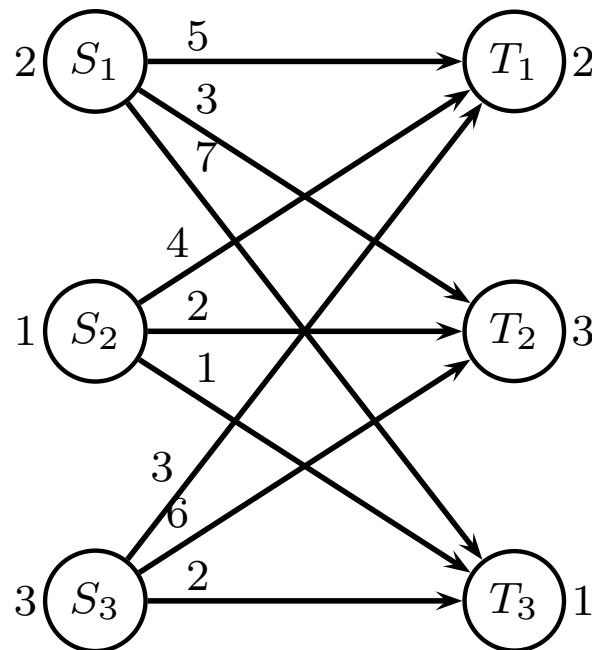
For instance:

- Columns are work plans, combine work plans to satisfy demand
- Columns are cutting patterns, combine patterns to satisfy demand
- Columns are cheapest paths, combine to flow through graph

In our case

- Stage 1: For each source S_i choose how the production should be flown
- Stage 2: Combine patterns, so that demand at T_j is satisfied and only one pattern per S_i is chosen.

Dantzig-Wolfe decomposition of problem



Fixed charge transportation problem

1 item sent from a S -node can be split in 3 ways (in our example, all legal if sent from S_2)

$$\begin{bmatrix} 1 & & \\ & 1 & \\ & & 1 \end{bmatrix} \Rightarrow \begin{bmatrix} 1 & & \\ & 1 & \\ & & 1 \end{bmatrix}$$

2 items from a S -node can be split in 6 ways (in our example 5 legal ways if sent from S_1)

$$\begin{bmatrix} 2 & & 1 & 1 & \\ & 2 & 1 & & 1 \\ & & 2 & 1 & 1 \end{bmatrix} \Rightarrow \begin{bmatrix} 2 & & 1 & 1 & \\ & 2 & 1 & & 1 \\ & & & 1 & 1 \end{bmatrix}$$

Fixed charge transportation problem

3 items sent from a S -node can be split in 10 ways (in our example 6 legal ways if sent from S_3)

$$\begin{bmatrix} 3 & & 2 & 2 & 1 & & 1 & & 1 \\ & 3 & & 1 & & 2 & 2 & & 1 & 1 \\ & & 3 & & 1 & & 1 & 2 & 2 & 1 \end{bmatrix} \Rightarrow \begin{bmatrix} & 2 & 2 & 1 & & 1 \\ 3 & 1 & & 2 & 2 & 1 \\ & & 1 & & 1 & 1 \end{bmatrix}$$

Fixed charge transportation problem

Variable $p_{ij} \in \{0, 1\}$ denotes if we choose pattern j for node S_i

Cost and pattern:

$$\begin{array}{rcl}
 & +18 \ p_{11} & +14 \ p_{12} \ +24 \ p_{13} \ +28 \ p_{14} \ +26 \ p_{15} \\
 & \text{-----} & \\
 S1: & + 2 \ p_{11} & + 0 \ p_{12} \ + 1 \ p_{13} \ + 1 \ p_{14} \ + 0 \ p_{15} \\
 & + 0 \ p_{11} & + 2 \ p_{12} \ + 1 \ p_{13} \ + 0 \ p_{14} \ + 1 \ p_{15} \\
 & + 0 \ p_{11} & + 0 \ p_{12} \ + 0 \ p_{13} \ + 1 \ p_{14} \ + 1 \ p_{15}
 \end{array}$$

$$\begin{array}{rcl}
 & +12 \ p_{21} & +10 \ p_{22} \ + 9 \ p_{23} \\
 & \text{-----} & \\
 S2: & + 1 \ p_{21} & + 0 \ p_{22} \ + 0 \ p_{23} \\
 & + 0 \ p_{21} & + 1 \ p_{22} \ + 0 \ p_{23} \\
 & + 0 \ p_{21} & + 0 \ p_{22} \ + 1 \ p_{23}
 \end{array}$$

$$\begin{array}{rcl}
 & +26 \ p_{31} & +28 \ p_{32} \ +24 \ p_{33} \ +31 \ p_{34} \ +30 \ p_{35} \ +35 \ p_{36} \\
 & \text{-----} & \\
 S3: & + 0 \ p_{31} & + 2 \ p_{32} \ + 2 \ p_{33} \ + 1 \ p_{34} \ + 0 \ p_{35} \ + 1 \ p_{36} \\
 & + 3 \ p_{31} & + 1 \ p_{32} \ + 0 \ p_{33} \ + 2 \ p_{34} \ + 2 \ p_{35} \ + 1 \ p_{36} \\
 & + 0 \ p_{31} & + 0 \ p_{32} \ + 1 \ p_{33} \ + 0 \ p_{34} \ + 1 \ p_{35} \ + 1 \ p_{36}
 \end{array}$$

Fixed charge transportation problem

Dantzig-Wolfe decomposed model, LP relaxed (**)

```
minimize
18p11+14p12+24p13+28p14+26p15+12p21+10p22+ 9p23+26p31+28p32+24p33+31p34+30p35+35p36
subject to
 2p11+ 0p12+ 1p13+ 1p14+ 0p15+ 1p21+ 0p22+ 0p23+ 0p31+ 2p32+ 2p33+ 1p34+ 0p35+ 1p36 = 2
 0p11+ 2p12+ 1p13+ 0p14+ 1p15+ 0p21+ 1p22+ 0p23+ 3p31+ 1p32+ 0p33+ 2p34+ 2p35+ 1p36 = 3
 0p11+ 0p12+ 0p13+ 1p14+ 1p15+ 0p21+ 0p22+ 1p23+ 0p31+ 0p32+ 1p33+ 0p34+ 1p35+ 1p36 = 1
 1p11+ 1p12+ 1p13+ 1p14+ 1p15                                     = 1
                                     + 1p21+ 1p22+ 1p23                                     = 1
                                     + 1p31+ 1p32+ 1p33+ 1p34+ 1p35+ 1p36 = 1
end
```

First 3 constraints: Meet demand at nodes T_j

Last 3 constraints: Only one patten for each S_i

Fixed charge transportation problem

Dantzig-Wolfe decomposed model, LP relaxed

```
CPLEX> opt
Dual simplex - Optimal:  Objective =  4.80000000000e+01
Solution time =      0.01 sec.  Iterations = 6 (0)
CPLEX> dis sol var -
Variable Name           Solution Value
p12                      1.000000
p22                      1.000000
p33                      1.000000
All other variables in the range 1-14 are 0.
```

) (not always an integer solution)

Fixed charge transportation problem

- Lower bound from Dantzig-Wolfe decomposed problem is much better
- Can reverse role of S and T to calculate other bound
- Choose best of bounds $S \rightarrow T$ and $T \rightarrow S$

Solving large problems

- If demands are large, impossible to write up Dantzig-Wolfe model
- *Delayed column generation*

Solving large LP problems

Frequently we have

$$\max cx$$

$$Ax = b$$

$$x \in \mathbb{R}_+^n$$

with thousands of constraints, and millions of variables.
However, columns in A follow some structure

Although Simplex in theory can solve these problems

- Problems can be too large to write up
- Problems cannot be stored in RAM on computer
- Numerical instability can make Simplex cycle

Solving large LP problems

For given basis B problem can be rewritten

$$-z + c_B x_B + c_N x_N = 0$$

$$A_B x_B + A_N x_N = b$$

Basis solution

$$-z + (c_N - c_B A_B^{-1} A_N) x_N = -c_B A_B^{-1} b$$

$$I x_B + A_B^{-1} A_N x_N = A_B^{-1} b$$

$$x_B, x_N \geq 0$$

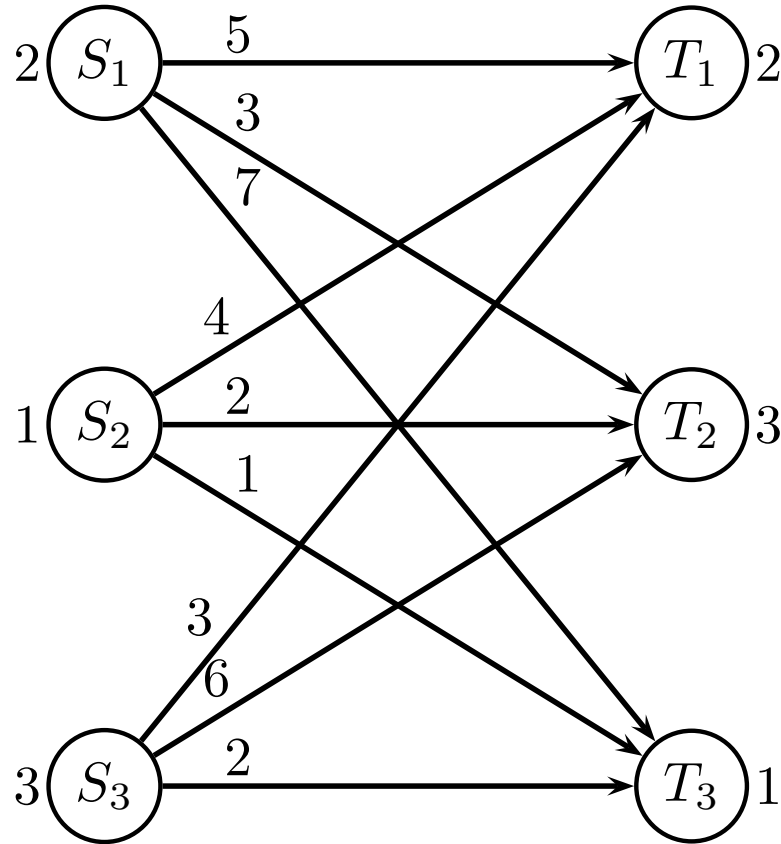
0				$c_N - y A_N$	$-y b$
1	0	...	0	$A_B^{-1} A_N$	$A_B^{-1} b$
0	1		0		
⋮	⋮	⋱	⋮		
0	0	...	1		

Solving large LP problems

- Assume number of columns much larger than number of constraints
- Simplex algorithm only needs basis B , columns A_B , reduced costs $c - yA$
- if we can provide this information, simplex algorithm does not need to know the full problem

Fixed charge transportation problem

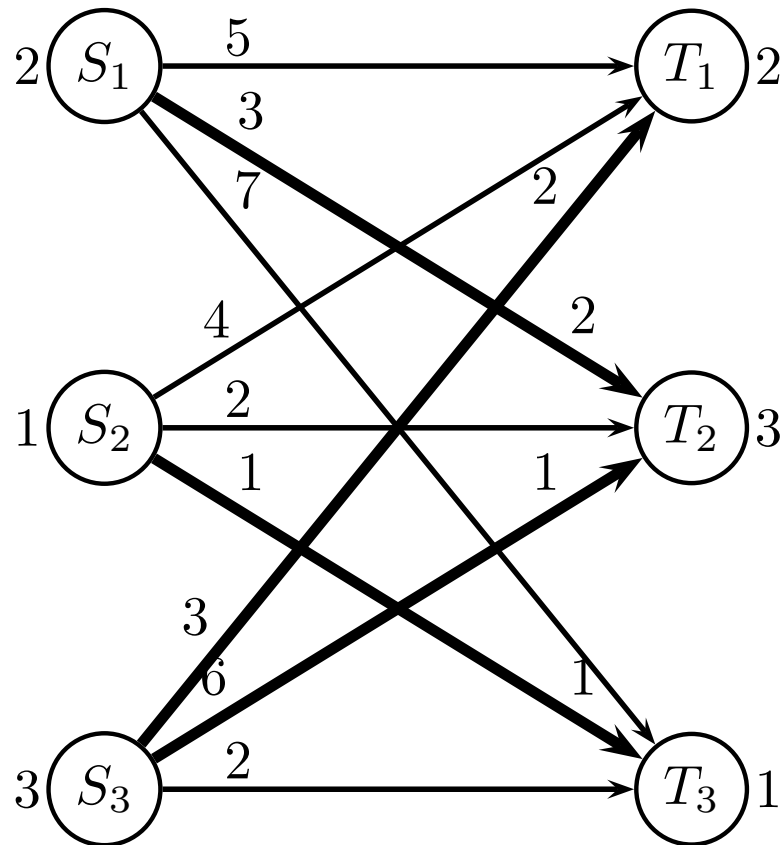
Greedy solution is found as last week (ignoring fixed charge)



(Normally fixed charge for using edge (i, j) is $F = 8$)

Fixed charge transportation problem

Greedy solution



Corresponds to patterns p_{12} , p_{23} and p_{32} .

Restricted master problem

Iteration 0: Master problem

```
minimize
    14p12+ 9p23+28p32
subject to
U1: 0p12+ 0p23+ 2p32 = 2
U2: 2p12+ 0p23+ 1p32 = 3
U3: 0p12+ 1p23+ 0p32 = 1
V1: 1p12                = 1
V2:      + 1p23          = 1
V3:      + 1p32          = 1
end
```

Solution: $z = 51$

Dual: $u_1 = 10.5, u_2 = 7, u_3 = 9$ and $v_1 = 0, v_2 = 0, v_3 = 0$

Reduced cost of a column

$$\bar{c} = c - yA$$

```

minimize
18p11+14p12+24p13+28p14+26p15+12p21+10p22+ 9p23+26p31+28p32+24p33+31p34+30p35+35p36
subject to
  2p11+ 0p12+ 1p13+ 1p14+ 0p15+ 1p21+ 0p22+ 0p23+ 0p31+ 2p32+ 2p33+ 1p34+ 0p35+ 1p36 = 2   (u1)
  0p11+ 2p12+ 1p13+ 0p14+ 1p15+ 0p21+ 1p22+ 0p23+ 3p31+ 1p32+ 0p33+ 2p34+ 2p35+ 1p36 = 3   (u2)
  0p11+ 0p12+ 0p13+ 1p14+ 1p15+ 0p21+ 0p22+ 1p23+ 0p31+ 0p32+ 1p33+ 0p34+ 1p35+ 1p36 = 1   (u3)
  1p11+ 1p12+ 1p13+ 1p14+ 1p15                                     = 1   (v1)
                                   + 1p21+ 1p22+ 1p23                                     = 1   (v2)
                                   + 1p31+ 1p32+ 1p33+ 1p34+ 1p35+ 1p36 = 1   (v3)
end

```

in our case

$$\bar{c}_{ij} = \left(\sum_j c_{ij} x_j + \sum_j F y_j \right) - \sum_j u_j x_j - v_i$$

Pricing problem

For each S_i we want the best splitting of the s_i items.
Let x_j denote how many items are sent from i to j .

$$\begin{aligned} \min \quad & \left(\sum_j c_{ij} x_j + \sum_j F y_j \right) - \sum_j u_j x_j - v_i \\ \text{s.t.} \quad & \sum_j x_j = s_i \\ & x_j \leq M y_j \\ & x_j \geq 0, \text{ integer} \\ & y_j \in \{0, 1\} \end{aligned}$$

The problem is a **multiple-choice knapsack problem**, solvable with dynamic programming

Solution x will define column in A matrix

Pricing problem S_1

For S_1 we get the model

```
minimize
    5 x1 + 3 x2 + 7 x3          (c_ij x_j)
    + 8 y1 + 8 y2 + 8 y3      (      F y_j)
    - 10.5 x1 - 7 x2 - 9 x3    ( u_j x_j)
    - 0                        ( v_i      )
subject to
    x1 + x2 + x3 = 2
    x1 <= 3 y1
    x2 <= 3 y2
    x3 <= 3 y3
binary
    y1 y2 y3
end
```

Solution:

$x_1 = 2, x_2 = 0, x_3 = 0$ with $z = -3$

Can be recognized as p_{11} .

Notice: for fixed y , constraint matrix is TU, so x integer

Pricing problem S_2

For S_2 we get the model

```
minimize
    4 x1 + 2 x2 + 1 x3
    + 8 y1 + 8 y2 + 8 y3
    - 10.5 x1 - 7 x2 - 9 x3
    - 0
subject to
    x1 + x2 + x3 = 1
    x1 <= 3 y1
    x2 <= 3 y2
    x3 <= 3 y3
binary
    y1 y2 y3
end
```

Solution:

$x_1 = 0, x_2 = 0, x_3 = 1$ with $z = 0$

Can be recognized as p_{23} .

Pricing problem S_3

For S_3 we get the model

```
minimize
    3 x1 + 6 x2 + 2 x3
    + 8 y1 + 8 y2 + 8 y3
    - 10.5 x1 - 7 x2 - 9 x3
    - 0
subject to
    x1 + x2 + x3 = 3
    x1 <= 3 y1
    x2 <= 3 y2
    x3 <= 3 y3
binary
    y1 y2 y3
end
```

Solution:

$x_1 = 2, x_2 = 0, x_3 = 1$ with $z = -6$

Can be recognized as p_{33} .

Pricing problem if columns known, S_1

2 items sent from S_1

	pattern x_j	$\sum_j c_{1j}x_j + \sum_j Fy_j$	$\sum_j u_jx_j$	v_1	red.cost
p_{11}	2 0 0	18	21	0	-3
p_{12}	0 2 0	14	14	0	0
p_{13}	1 1 0	24	17.5	0	6.5
p_{14}	1 0 1	28	20.5	0	7.5
p_{15}	0 1 1	26	16	0	10

$$u_1 = 10.5, u_2 = 7, u_3 = 9 \text{ and } v_1 = 0, v_2 = 0, v_3 = 0$$

● Notice that $\sum_j c_{1j}x_j + \sum_j Fy_j$ is “old” cost of pattern

Pricing problem if columns known, S_2

1 item sent from S_2

	pattern x_j	$\sum_j c_{2j}x_j + \sum_j Fy_j$	$\sum_j u_jx_j$	v_2	red.cost
p_{21}	1 0 0	12	10.5	0	1.5
p_{22}	0 1 0	10	7	0	3
p_{23}	0 0 1	9	9	0	0

$$u_1 = 10.5, u_2 = 7, u_3 = 9 \text{ and } v_1 = 0, v_2 = 0, v_3 = 0$$

Pricing problem if columns known, S_3

3 items sent from S_3

	pattern x_j	$\sum_j c_{3j}x_j + \sum_j Fy_j$	$\sum_j u_jx_j$	v_3	red.cost
p_{31}	0 3 0	26	21	0	5
p_{32}	2 1 0	28	28	0	0
p_{33}	2 0 1	24	30	0	-6
p_{34}	1 2 0	31	24.5	0	6.5
p_{35}	0 2 1	30	23	0	7
p_{36}	1 1 1	35	26.5	0	8.5

$$u_1 = 10.5, u_2 = 7, u_3 = 9 \text{ and } v_1 = 0, v_2 = 0, v_3 = 0$$

Restricted master problem

Iteration 1: Adding pattern p_{33} to master problem

```
minimize
    14p12+ 9p23+28p32+24p33
subject to
U1: 0p12+ 0p23+ 2p32+ 2p33 = 2
U2: 2p12+ 0p23+ 1p32+ 0p33 = 3
U3: 0p12+ 1p23+ 0p32+ 1p33 = 1
V1: 1p12                      = 1
V2:      + 1p23                = 1
V3:      + 1p32+ 1p33          = 1
end
```

Solution: $z = 51$

Dual: $u_1 = 14, u_2 = 0, u_3 = -4$ and $v_1 = 14, v_2 = 13, v_3 = 0$

Pricing problem S_1

For S_1 we get the model

```
minimize
    5 x1 + 3 x2 + 7 x3
    + 8 y1 + 8 y2 + 8 y3
    - 14 x1 - 0 x2 + 4 x3
    - 14
subject to
    x1 + x2 + x3 = 2
    x1 <= 3 y1
    x2 <= 3 y2
    x3 <= 3 y3
binary
    y1 y2 y3
end
```

Solution:

$x_1 = 2, x_2 = 0, x_3 = 0$ with $z = -24$

Can be recognized as p_{11} .

Pricing problem S_2

For S_2 we get the model

```
minimize
    4 x1 + 2 x2 + 1 x3
    + 8 y1 + 8 y2 + 8 y3
    - 14 x1 - 0 x2 + 4 x3
    - 13
subject to
    x1 + x2 + x3 = 1
    x1 <= 3 y1
    x2 <= 3 y2
    x3 <= 3 y3
binary
    y1 y2 y3
end
```

Solution:

$x_1 = 1, x_2 = 0, x_3 = 0$ with $z = -15$

Can be recognized as p_{21} .

Pricing problem S_3

For S_3 we get the model

```
minimize
    3 x1 + 6 x2 + 2 x3
    + 8 y1 + 8 y2 + 8 y3
    - 14 x1 - 0 x2 + 4 x3
    - 0
subject to
    x1 + x2 + x3 = 3
    x1 <= 3 y1
    x2 <= 3 y2
    x3 <= 3 y3
binary
    y1 y2 y3
end
```

Solution:

$x_1 = 2, x_2 = 0, x_3 = 1$ with $z = 0$

Can be recognized as p_{33} .

Restricted master problem

Iteration 2: Adding pattern p_{11} to master problem

```
minimize
    18p11+14p12+ 9p23+28p32+24p33
subject to
U1: 2p11+ 0p12+ 0p23+ 2p32+ 2p33 = 2
U2: 0p11+ 2p12+ 0p23+ 1p32+ 0p33 = 3
U3: 0p11+ 0p12+ 1p23+ 0p32+ 1p33 = 1
V1: 1p11+ 1p12                        = 1
V2:                + 1p23                = 1
V3:                + 1p32+ 1p33 = 1
end
```

Solution: $z = 51$

Dual: $u_1 = 0, u_2 = 7, u_3 = 0$ and $v_1 = 0, v_2 = 9, v_3 = 21$

Pricing problem, S_1, S_2, S_3

Iteration 2: Pricing problem

For S_1 : $z \geq 0$

For S_2 : $x_1 = 0, x_2 = 1, x_3 = 0$ with $z = -6$ (pattern p_{22})

For S_3 : $x_1 = 2, x_2 = 0, x_3 = 1$ with $z = -16$ (pattern p_{31})

Restricted master problem

Iteration 3: Adding pattern p_{31} to master problem

```
minimize
    18p11+14p12+ 9p23+26p31+28p32+24p33
subject to
U1: 2p11+ 0p12+ 0p23+ 0p31+ 2p32+ 2p33 = 2
U2: 0p11+ 2p12+ 0p23+ 3p31+ 1p32+ 0p33 = 3
U3: 0p11+ 0p12+ 1p23+ 0p31+ 0p32+ 1p33 = 1
V1: 1p11+ 1p12                                = 1
V2:                + 1p23                        = 1
V3:                + 1p31+ 1p32+ 1p33 = 1
end
```

Solution: $z = 51$

Dual: $u_1 = 1, u_2 = 0, u_3 = -4$ and $v_1 = 14, v_2 = 13, v_3 = 26$

Pricing problem, S_1, S_2, S_3

Iteration 3: Pricing problem

For S_1 : $z \geq 0$

For S_2 : $x_1 = 0, x_2 = 1, x_3 = 0$ with $z = -3$ (pattern p_{22})

For S_3 : $z \geq 0$

Restricted master problem

Iteration 4: Adding pattern p_{22} to master problem

```
minimize
    18p11+14p12+10p22+ 9p23+26p31+28p32+24p33
subject to
U1: 2p11+ 0p12+ 0p22+ 0p23+ 0p31+ 2p32+ 2p33 = 2
U2: 0p11+ 2p12+ 1p22+ 0p23+ 3p31+ 1p32+ 0p33 = 3
U3: 0p11+ 0p12+ 0p22+ 1p23+ 0p31+ 0p32+ 1p33 = 1
V1: 1p11+ 1p12                                     = 1
V2:           + 1p22+ 1p23                             = 1
V3:           + 1p31+ 1p32+ 1p33 = 1
end
```

Solution: $z = 48, p_{12} = p_{22} = p_{33} = 1$

Dual: $u_1 = 8.16, u_2 = 8.66, u_3 = 7.66$ and
 $v_1 = -3.33, v_2 = 1.33, v_3 = 0$

Pricing problem, S_1, S_2, S_3

Iteration 4: Pricing problem

For S_1 : $z \geq 0$

For S_2 : $z \geq 0$

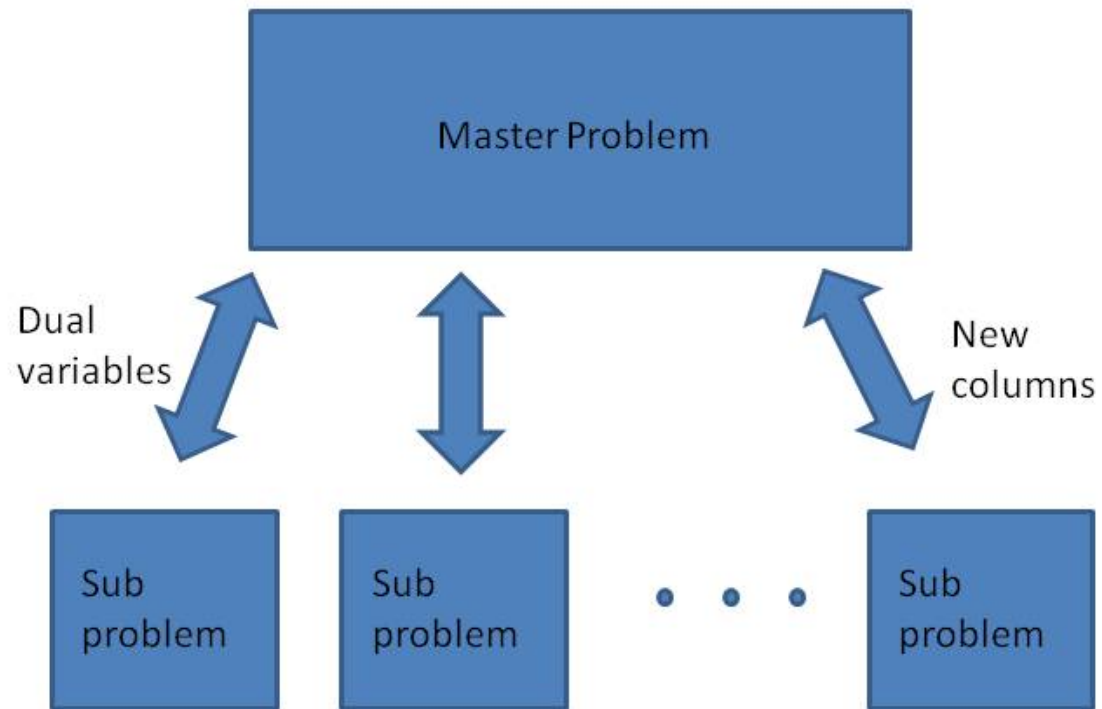
For S_3 : $z \geq 0$

Reduced costs are positive, so **stop**

Fixed charge transportation problem

- Only 4 iterations (7 columns out of 14 generated)
- Much better lower bounds than simple formulation
- Pricing problem quite easy to solve
- Objective function z moves in large “jumps”
- Using “ $\geq$ ” constraints in master may give more smooth development of objective
($z = 51, 51, 51, 49.7, 48$, paths added $p_{21}, p_{33}, p_{31}, p_{22}$)
Dual variables are restricted to ≥ 0 , so cannot fluctuate so much (deep dual inequalities)

Column generation



Column generation

- Find initial columns
- Repeat
 - Solve restricted master problem (LP-problem)
 - Find dual prices
 - Solve pricing problem(s)
 - Add new column(s)
- Until no columns with negative reduced cost

This solves the LP-problem to optimality
(but often IP problem also needs to be solved)

Tips-and-tricks

Column generation for fixed-charge transport

- check that problem is balanced, otherwise add surplus node to make problem balanced
- if some edges are missing, you can add them with big cost M . This will ensure that you can find a greedy solution
- You need to use an LP-solver to find dual variables. You cannot use the method from last week (it is a different LP-model)
- if you generate the same column twice, then there is something wrong
- avoid upper bounds on variables $0 \leq x_j \leq 1$ since they have an associated dual variable

Appendix: Pricing problem solved fast

Consider node S_i with production s_i (or just s)

Let x_{jk} indicate if we send k units to node T_j

multiple choice knapsack, capacity s

$$\min \left\{ \sum_{j=1}^n \sum_{k=1}^s c_{jk} x_{jk} \left| \sum_{j=1}^n \sum_{k=1}^s k \cdot x_{jk} = s \right. \right\}$$

Subproblem for $0 \leq a \leq s$

$$f_i(a) = \min \left\{ \sum_{j=1}^i \sum_{k=1}^a c_{jk} x_{jk} \left| \sum_{j=1}^i \sum_{k=1}^a k \cdot x_{jk} = a \right. \right\}$$

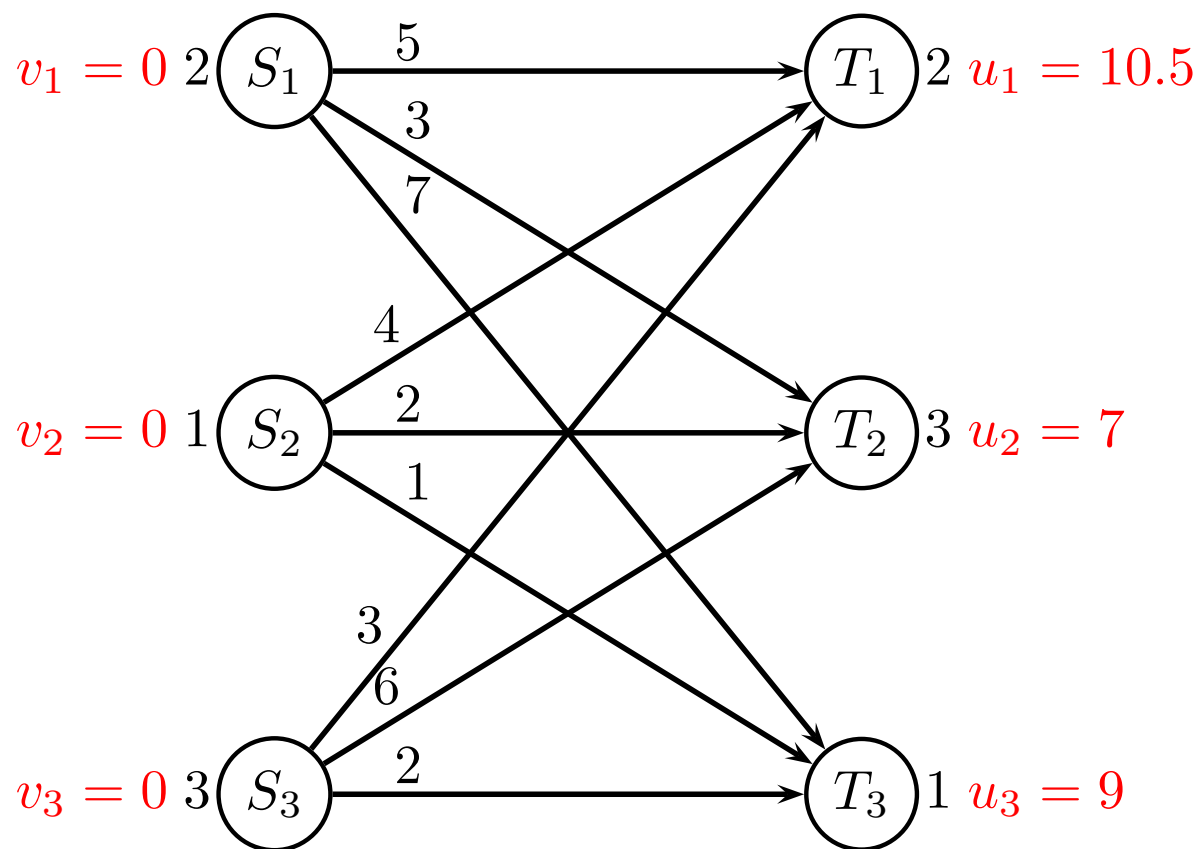
Recursion

$$f_i(a) = \min_{k=0, \dots, a} \{ f_{i-1}(a - k) + c_{ik} \}$$

Start condition

$$f_1(a) = c_{1a} \quad \forall a$$

Pricing problem, multiple-choice knapsack



For node S_1 we have pairs (c_{jk}, k) for $k = 0, 1, 2$

$$T_1 \begin{pmatrix} 0 + 0 & 0 \\ -5.5 + 8 & 1 \\ -11 + 8 & 2 \end{pmatrix} \quad T_2 \begin{pmatrix} 0 + 0 & 0 \\ -4 + 8 & 1 \\ -8 + 8 & 2 \end{pmatrix} \quad T_3 \begin{pmatrix} 0 + 0 & 0 \\ -2 + 8 & 1 \\ \infty & 2 \end{pmatrix}$$

Pricing problem, multiple-choice knapsack

For node S_1 we have costs c_{jk}

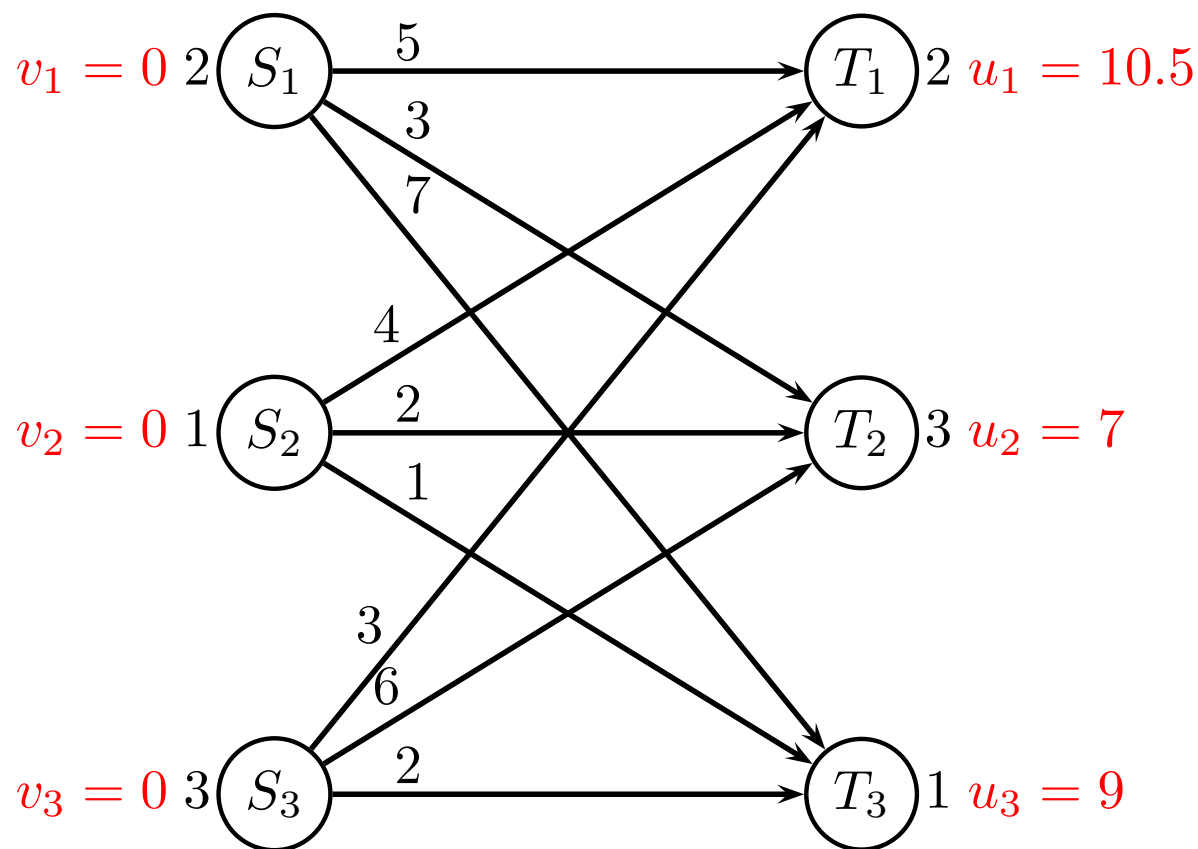
$k \backslash j$	1	2	3
0	0	0	0
1	2.5	4	6
2	-3	0	∞

Solving dynamic programming

$a \backslash i$	1	2	3
0	0	0	0
1	2.5	2.5	2.5
2	-3	-3	-3

Optimal solution $z = -3$ pattern 2,0,0

Pricing problem, multiple-choice knapsack



For node S_2 we have pairs (c_{jk}, k) for $k = 0, 1$

$$T_1 \begin{pmatrix} 0 + 0 & 0 \\ -6.5 + 8 & 1 \end{pmatrix} \quad T_2 \begin{pmatrix} 0 + 0 & 0 \\ -5 + 8 & 1 \end{pmatrix} \quad T_3 \begin{pmatrix} 0 + 0 & 0 \\ -8 + 8 & 1 \end{pmatrix}$$

Pricing problem, multiple-choice knapsack

For node S_2 we have costs c_{jk}

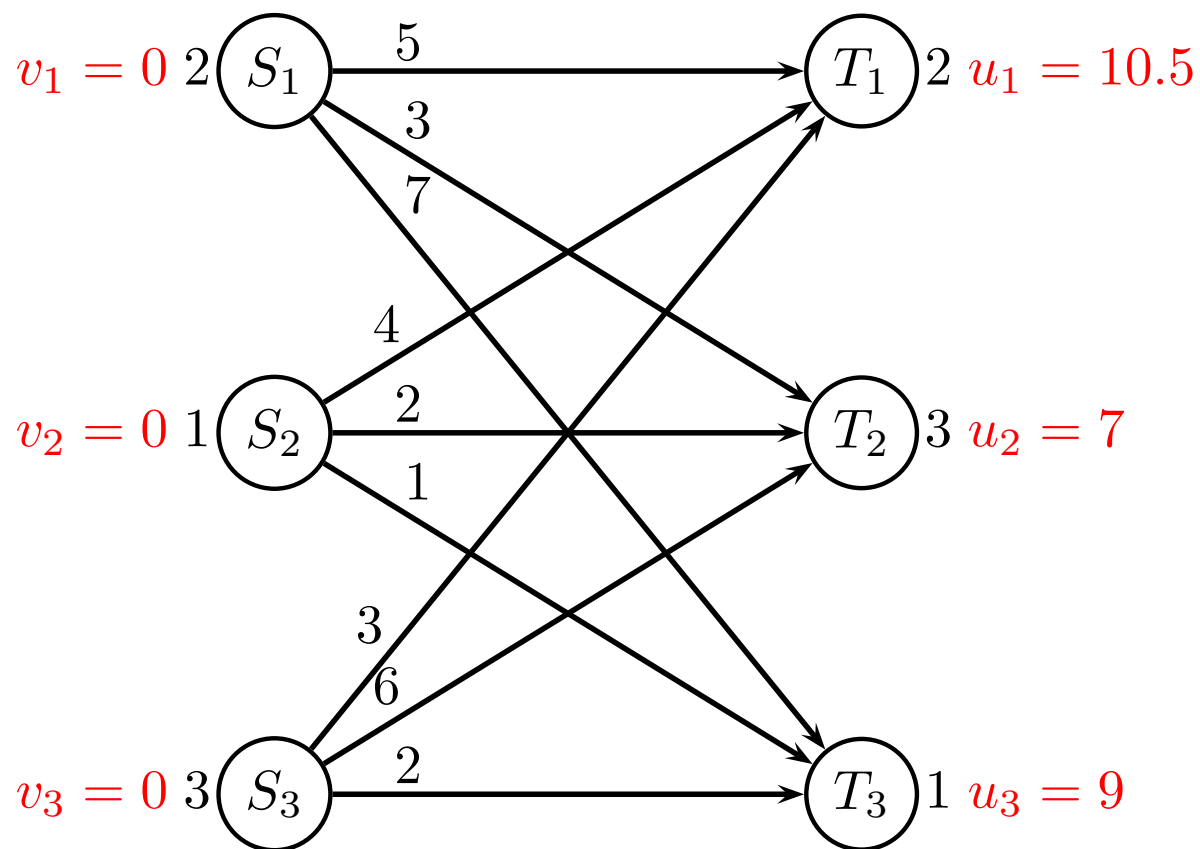
$k \backslash j$	1	2	3
0	0	0	0
1	1.5	3	0

Solving dynamic programming

$a \backslash i$	1	2	3
0	0	0	0
1	1.5	1.5	0

Optimal solution $z = 0$ pattern 0,0,1

Pricing problem, multiple-choice knapsack



For node S_3 we have pairs (c_{jk}, k) for $k = 0, 1, 2, 3$

$$T_1 \begin{pmatrix} 0 + 0 & 0 \\ -7.5 + 8 & 1 \\ -15 + 8 & 2 \\ \infty & 3 \end{pmatrix} \quad T_2 \begin{pmatrix} 0 + 0 & 0 \\ -1 + 8 & 1 \\ -2 + 8 & 2 \\ -3 + 8 & 3 \end{pmatrix} \quad T_3 \begin{pmatrix} 0 + 0 & 0 \\ -7 + 8 & 1 \\ \infty & 2 \\ \infty & 3 \end{pmatrix}$$

Pricing problem, multiple-choice knapsack

For node S_3 we have costs c_{jk}

$k \backslash j$	1	2	3
0	0	0	0
1	0.5	7	1
2	-7	6	∞
3	∞	5	∞

Solving dynamic programming

$a \backslash i$	1	2	3
0	0	0	0
1	0.5	0.5	0.5
2	-7	-7	-7
3	∞	0	-6

Optimal solution $z = -6$ pattern 2,0,1